

● Environment setup:

1) Details of *requirements* files:

```
a) scikit-learn  
b) tensorflow  
c) matplotlib  
d) numpy  
e) pandas  
f) yahoo_fin  
g) yfinance
```

I had to change sklearn from the original file to scikit-learn.

Also p1 uses yfinance while v0.1 uses yahoo_fin

2) I dealt with the following errors:

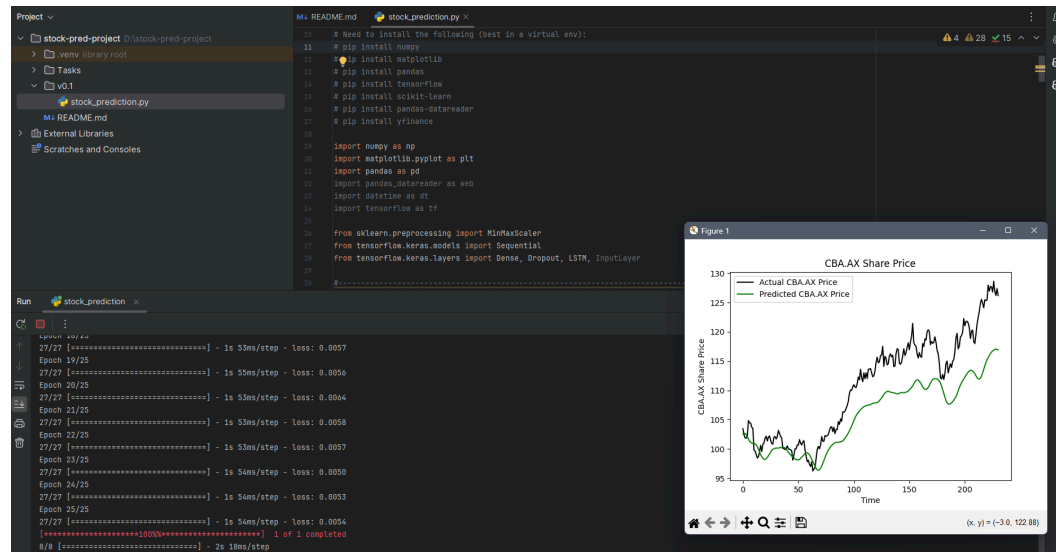
a) I was trying to run stock_prediction.py in p1, where I was supposed to run train.py

```
Warning - Certain functionality  
requires requests_html, which is not installed.  
  
Install using:  
pip install requests_html  
  
After installation, you may have to restart your Python session.
```

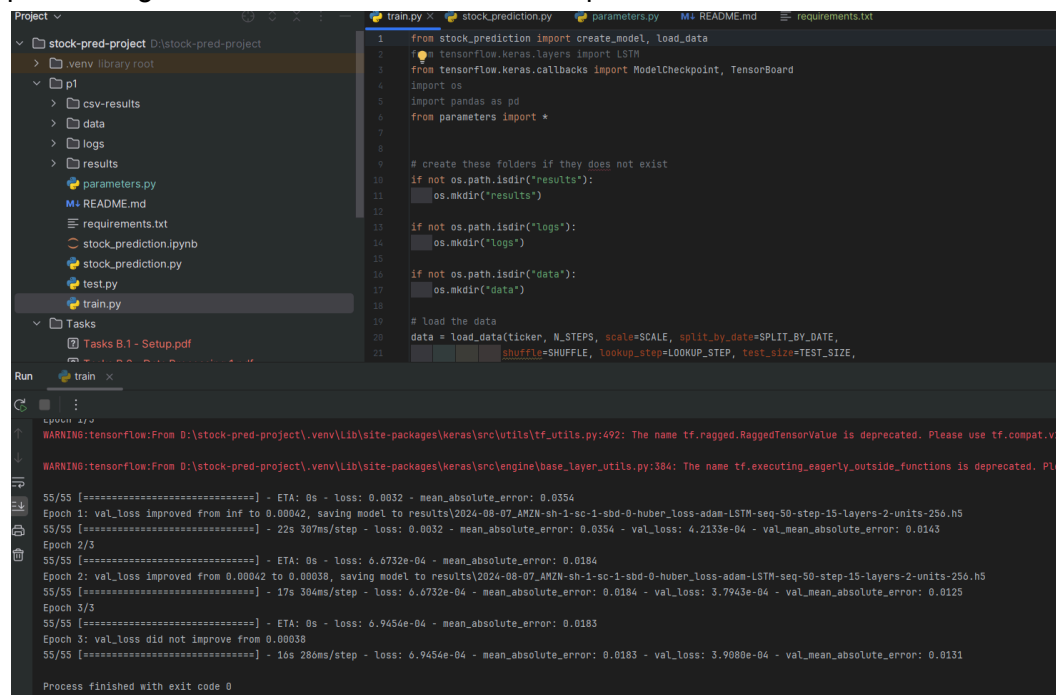
b) In v0.1, I had the *batch_input_shape* issue. Solved it by installing tensorflow 2.15.0.

● Testing the provided code:

1) v0.1:



2) p1: running with EPOCH = 3 for ease of setup



• Summary of the code:

The codebase v.01 uses *yfinance* to get the stock data from *TRAIN_START* to *TRAIN_END*. The model works on the *CLOSE* price of the stock. The model stores training and test data as numpy array as they are convenient to deal with.

- It uses scikit-learn MinMaxscaler to bring all feature values between 0 and 1.

- It reshapes the feature data array as a 2d array as that is required for the scaler.
- After scaling, it flattens the array back to 1d.
- Then it runs a for loop from range(prediction days = 60 to end of data), so it selects different batches of size 60 from the data for training.
- Then it converts the data to a np array and reshapes it to a 3d array with structure(number of data points, number of time steps, number of features per timestep = 1)
- Then it uses the model “sequential”
- Then it adds a LSTM layer with 50 neurons with input shape = (number of time_steps in training data, 1) and sets the return sequences as true so that we can use stacked LSTM.
- To avoid overfitting, it sets a dropout layer of 20% frequency, which removes 20 units at random after each lstm laer.
- It has 3 LSTM layers in total. At the end, ti adds a Dense layer for the output.
- Then the model uses adam optimiser with mean_squared_loss.
- Then we move on to testing the data: we include test data and last 60 days prediction values.
- Then reshaping and scaling and selecting batches of length 60 for testing.
- Then we plot the graphs.
- Additional: Plot candlestick charts with tracer for next day prediction.

Candlestick chart:

